

Acknowledgment

I would like to express my sincere gratitude to **Elin Electronics Ltd.** for providing me with the opportunity to undertake my internship and gain practical exposure to industrial processes and control systems. This internship has been an invaluable learning experience, allowing me to bridge the gap between theoretical concepts and real-world engineering applications.

I am especially grateful to **Mr. Naresh Kumar**, my internship supervisor at **Elin Electronics Ltd.**, for his valuable guidance, continuous support, and encouragement throughout the internship. His technical insights and constructive suggestions greatly contributed to the successful completion of my project.

I am deeply thankful to the management and technical staff of **Elin Electronics Ltd.** for their guidance, encouragement, and support throughout the internship period. Their valuable insights into manufacturing processes and industrial automation greatly enhanced my understanding of control engineering and its practical applications.

I would also like to express my heartfelt gratitude to the faculty members of the **Department of Instrumentation and Control Engineering, Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar**, for providing me with the academic foundation and motivation to undertake this project. Their continuous encouragement has played a significant role in the successful completion of my internship.

Finally, I would like to thank my family and friends for their constant support, motivation, and encouragement throughout this internship. Their confidence in me inspired me to complete this project successfully.

Table of Contents

- 1. Introduction**
- 2. Objective of Internship**
- 3. Company Profile**
 - i. About Elin Electronics
 - ii. Organizational Overview
 - iii. Products and Services
- 4. Project Focus Area**
 - i. Reflow Oven
 - ii. Surface Mount Technology (SMT)
 - iii. Multi-Zone Temperature Control
- 5. Internship Roles and Responsibilities**
- 6. Project: Multi-Zone Reflow Oven Temperature Control using PI Controller**
 - i. Problem Statement
 - ii. Literature Overview and Design Justification
 - iii. System Modelling
 - iv. Transfer Function Development
 - v. PI Controller Design
 - vi. Temperature Profile Tracking
 - vii. Time Constant Comparison
 - viii. Disturbance Analysis
 - ix. Manual PI Tuning
 - x. Automatic PI Tuning
 - xi. Dashboard Implementation
 - xii. Simulation Results and Discussion
- 7. Challenges Faced**
- 8. Conclusion**
- 9. Future Scope**
- 10. References**
- 11. Internship Certificate**

1.Introduction

The rapid advancement in electronics technology has led to the significant increase in demand for a compact, reliable and high performance of electronic devices. Nowadays, Surface Mount Technology (SMT) has become the standard manufacturing process for assembling the modern electronic circuits due to its high production speed, improved reliability and efficient utilization of printed circuit boards (PCBs). SMT process involves a crucial stage of reflow soldering, in which the electronic components are permanently gets attached to the PCB using a controlled heating process.

A reflow oven is a specialized industrial machine which is designed to heat assembled PCBs according to a predefined temperature profile. The profile consists of 4 stages which are as following-

- Preheating
- Soaking
- Reflow
- And Cooling

Each stage plays a significant role, ensuring proper solder melting, strong electrical connections and prevents the damage of components. And if the temperature becomes higher or falls below the desired value, then the soldering effects such as cold joints, solder bridges, insufficient wetting, or component failure may occur. Therefore, a precise temperature regulation is one of the most critical requirements in the reflow soldering process.



Figure 1.1: Industrial Multi-Zone Reflow Oven.

Generally, modern reflow ovens consists of multiple heating zones, and each zone is responsible for maintaining a specific range of temperature. Accurate control of these heating zones ensures that the PCB follows the desired thermal profile throughout the soldering process. To achieve this, automatic control systems are employed to continuously monitoring the temperature and adjust the heater output correctly.

There are various control techniques, but the Proportional-Integral (PI) controller is widely used in industrial thermal system because of its simplicity, reliability and ability to minimize steady state error while maintaining stable system performance. Proper tuning of the PI controller improves the response of heating system, reduces temperature fluctuations and enhances the overall quality of soldering process.

In this internship project I have modelled a simplified multi-zone reflow oven temperature control system and simulated using the MATLAB/ Simulink. I have used a first order transfer function to represent the thermal dynamics of the heating system. I performed both manual tuning and automatic tuning using MATLAB's `pidentune()` function to obtain a suitable PI controller parameters. The performance of the controller was analysed under different operating conditions including the variations in the time constant of system and introduction of externally applied disturbances. I have also added dashboard consisting of a circular gauge, numeric display and an indicator lamp to visualize the simulated oven temperature in real time.

The objective of this project is to understand the application of control engineering principles in industrial thermal systems and to evaluate the effectiveness of PI control in maintaining the desired temperature profile of a reflow oven. The knowledge that I have gained through this work gave me a practical insight into process control, system modelling, controller design and simulation techniques used in modern electronics manufacturing.

2. Objective of Internship

The primary objective of this internship was to gain a practical exposure to industrial control systems and their applications in electronics manufacturing. This internship was focused to understand the working principles of reflow ovens used in Surface Mount Technology (SMT) and the importance of accurate temperature control during the soldering process.

The major objective of this internship are as follows:

- To understand the working principle and industrial applications off reflow ovens in SMT manufacturing.
- To study the significance of temperature profile tracking to achieve high quality solder joints.
- To develop a mathematical model of a reflow oven heating system using MATLAB/Simulink.
- To design and implementing a Proportional-Integral (PI) controller to regulate the temperature of oven.
- To perform both manual and automatic controller using MATLAB's pidtune() function.
- To analyse the system performance when introduced different time constants and external disturbances.
- To develop a simple monitoring dashboard containing a circular gauge, numerical display and a temperature indicator lamp.
- To enhance the practical knowledge of process control, system modelling, controller design and simulation techniques used in industrial automation.

3. Company Profile

3.1 About Elin Electronics Ltd.

Elin Electronics is India's one of the leading companies among the Electronics Manufacturing Services (EMS) companies, providing end to end manufacturing solutions for a wide range of electronics products. The Elin Group was founded in 1969 in Ghaziabad and throughout the time the company has developed expertise in designing, manufacturing and assembling electronic products for both domestic and international markets. With multiple manufacturing facilities across India, Elin Electronics serves customers from various sectors including consumer electronics, home appliances, lighting, automotive electronics and medical equipment.

The company is specialized in high quality electronic manufacturing by integrating advanced production techniques, automated assembly lines and stringent quality control procedures. Its focus on innovation, precision and customer satisfaction has made it a trusted manufacturer partner with several well-known partners.

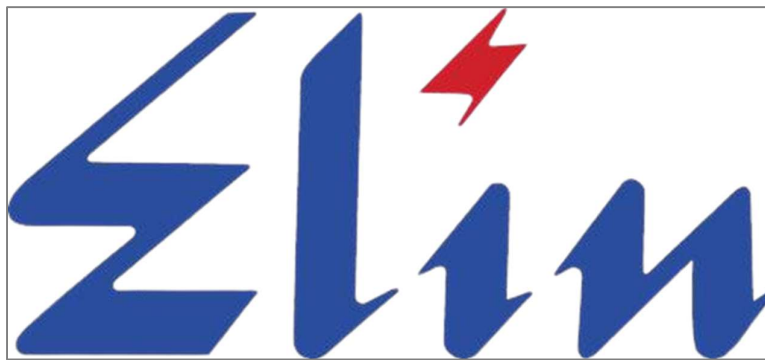


Figure 3.1: Corporate Logo of Elin Electronics Limited.

3.2 Organizational Overview

Elin Electronics follows a well-structured organizational framework that enables an efficient coordination between various departments involving product development and manufacturing. The company consists of specialized teams which are responsible for research

and development , production, quality assurance, supply chain management, testing, maintenance and customer support.

Modern manufacturing facilities integrated with automated production lines, inspection systems and testing equipment ensures a consistent product quality and high production efficiency. Continuous improvement in processes and adherence to international quality standards contributes significantly to the company's operational excellence.

3.3 Products and Services

Elin Electronics manufactures and assembles a diverse range of electronic products and provides comprehensive Electronic Manufacturing Services (EMS). Some of the major products and service includes:

- Consumer electronic products
- LED lighting solutions
- Home appliance electronics
- Automotive electronic assemblies
- Medical electronic devices
- Printed Circuit Board Assemblies (PCBAs)
- Original Equipment Manufacturing (OEM)
- Original Design Manufacturing (ODM)

The company also offers PCB assembly, testing, quality inspection, product integration and complete manufacturing support for various industries.

4. Project Focus Area

This part of the report introduces the technologies and concepts used in the project therefore before explaining the actual implementation the heading is divided into four sub sections for a better understanding.

4.1 Reflow Oven

A reflow oven is the heating system used in industries in the Surface Mount Technology (SMT) for soldering the electronic components on Printed Circuit Boards (PCBs). When the electronic components are placed on a PCB using a solder paste, after that the assembled board passes through a reflow oven where it is allowed to heat according to a carefully controlled temperature profile. And this controlled heating process melts the solder paste, and thus forming a reliable electrical and mechanical connections between the components and the PCB.

Nowadays, modern reflow ovens consists of multiple heating zones, thus each operating at different temperatures to gradually heat and cool the PCB. To produce a high-quality solder joint and preventing the electronic components from damage, maintenance of correct temperature is an essential point. Thus, accurate temperature control plays crucial role in ensuring a reliable manufacturing and product quality.

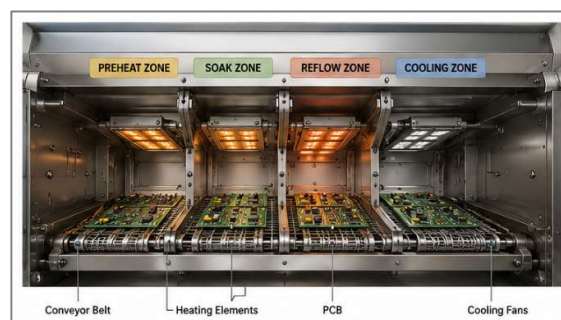


Figure 4.1 Internal view of a Multi-zone reflow oven

4.2 Surface Mount Technology

Surface Mount Technology (SMT) is a widely used manufacturing process in which electronic components such as resistor, capacitor, inductor and logic gates are mounted directly on surface

of Printed Circuit Boards (PCBs). Traditional method involves the hole technology, in which components are fitted in the holes drilled on the PCBs. The hole technology has drawbacks of time consumption, more surface area used and it is technically difficult to drill such a small hole on circuit boards. That's why SMT is more reliable than the hole technology. Unlike traditional method, SMT allows components to be placed using high speed pick and place machines, and results in a compact circuit designs, higher production efficiency and improved reliability.

The SMT manufacturing process follows the following stages:



Among all these stages, reflow soldering is one of the most critical processes because the quality of solder joints directly affects the reliability and performance of the final electronic product.



Figure 4.2 SMT Manufacturing Process

4.3 Multi Zone Temperature Control

Modern reflow ovens contain multiple heating zones that regulate the temperature at different stages of soldering process independently. Each zone performs a specific function such as

preheating, soaking, reflow or cooling to ensure that the PCB follows the required thermal profile.

Although this internship project models the thermal system using a simplified first order transfer function, the developed PI controller demonstrates the fundamental principles of temperature regulation used in industrial multi zone reflow ovens. The simulation provides a valuable insight into controller design, system response and temperature profile tracking in thermal process control.

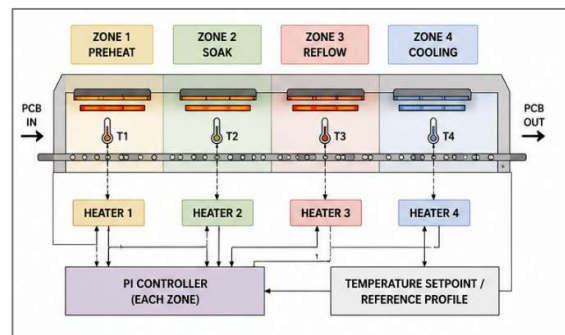


Figure 4.3: Multi-Zones of Reflow Oven.

5. Internship Role and Responsibilities

During the internship, I worked on a simulation-based project titled “**Multi Zone Reflow Oven Temperature Control Using a PI Controller**”. My primary responsibility was to understand the temperature control mechanism used in reflow soldering and develop a MATLAB Simulink model to simulate the thermal behaviour of a multi zone reflow oven.

The major roles and responsibilities taken during the internship are as follows:

- Studied the principle of reflow ovens used in Surface Mount Technology (SMT).
- Understood the importance of temperature profile tracking during the soldering process.
- Developed a mathematical model of the thermal system using a first order transfer function.
- Designed and implemented a PI controller in MATLAB/Simulink for temperature regulation.
- Performed manual tuning and automatic tuning of the PI controller.
- Simulated the system under different time constants to study its dynamic response.
- Introduced external disturbances to analyse the robustness of the controller.
- Developed a monitoring dashboard consisting of a circular gauge, numerical display, and indicator lamp for real time visualization.
- Recorded simulation results, compared system responses and documented the project using GitHub and technical documentation.

6. Project: Multi-Zone Reflow Oven Temperature Control Using PI Controller

6.1 Problem Statement

The objective of this project is developing a simulation model of a multi zone reflow oven temperature control system which should be capable of tracking a predefined temperature profile accurately by using a Proportional-Integral (PI) controller. In industries, it is essential to maintain the desired thermal profile to produce reliable solder joints and ensuring the quality of printed circuit board assemblies. However, thermal systems exhibit slow dynamic responses and are susceptible to variations in operating conditions, making precise temperature regulation a challenging task.

Since direct access to an industrial reflow oven was not available to me during the internship, that's why I developed a virtual model in MATLAB/Simulink to study the thermal behaviour of the system. This project focuses on designing an effective PI controller, evaluating its performance under different thermal time constants and analysing its response to external disturbances. A monitoring dashboard was also developed to provide a real time visualization of system output during simulation.

6.2 Literature Overview and Design Justification

Temperature regulation in industrial heating systems has been extensively studied due to its significant impact on manufacturing quality and process reliability. There are various control techniques that have been employed for thermal process control, each offers different levels of complexity and performance depending on the application requirements.

The simplest and easiest approach is On-Off Control, which switches the heater between fully On and Off states alternately. Although easy to implement, this method results in large temperature oscillations and is therefore becomes unsuitable for applications that requires precise thermal regulation.

To improve control performance, Proportional-Integral(PI) and Proportional-Integral-Derivative(PID) controllers are used in industries. PI controllers eliminate the steady state error while maintaining a simple control structure relatively and making them well suited for slow

thermal system such as industrial ovens. Although PID controllers include derivative action as well to improve transient response, the derivative term amplifies the measurement noise and provides a limited additional benefit in thermal applications where system dynamics are inherently slow.

The more advanced control techniques include Fuzzy Logic Control and Model Predictive Control (MPC). These strategies also have been investigated for reflow oven temperature control and it is found that they can offer improved performance under varying operating conditions but they generally require a more complex system modelling, computational resources and implementation efforts.

Considering the objectives of this internship project, a PI controller was selected because it provides an effective balance between control performance, implementation simplicity and industrial applicability. MATLAB/Simulink was chosen as the development platform because of its powerful simulation capabilities, controller design tools and ease of analysing system responses under different operating conditions before practical implementation.

6.3 System Modelling

To study the temperature regulation of a multi zone reflow oven, a simulation model was developed using the MATLAB/Simulink. The objective of the model was to replicate the thermal behaviour of reflow oven and evaluate the performance of a PI controller under different operating conditions. Since the project was simulation based, the model provides a safe and cost-effective environment for analysing controller performance without the need of physical hardware.

The system was designed as a closed loop feedback control system, in which the desired temperature profile acts as the reference input. The actual output temperature is measured continuously and compared with reference value. The difference between the reference temperature and measured temperature is called the error signal and is supplied to the PI controller. Based on this error the controller generates an appropriate control signal to regulate the heating process and minimizes the temperature deviation.

The thermal characteristics of the reflow oven are represented using a first order transfer function which approximate the heating dynamics of the system. A disturbance signal was

introduced into the system to simulate the unexpected variations in the heating process. Thus, the model was evaluated under a practical condition. System response was monitored using a scope block, while a dash board consisting of a circular gauge, numeric display and an indicator lamp was developed for a real time visualization of the simulated model.

The complete Simulink model forms the foundation for subsequent analyses, including controller tuning, temperature profile tracking, time constant comparison and disturbance rejection.

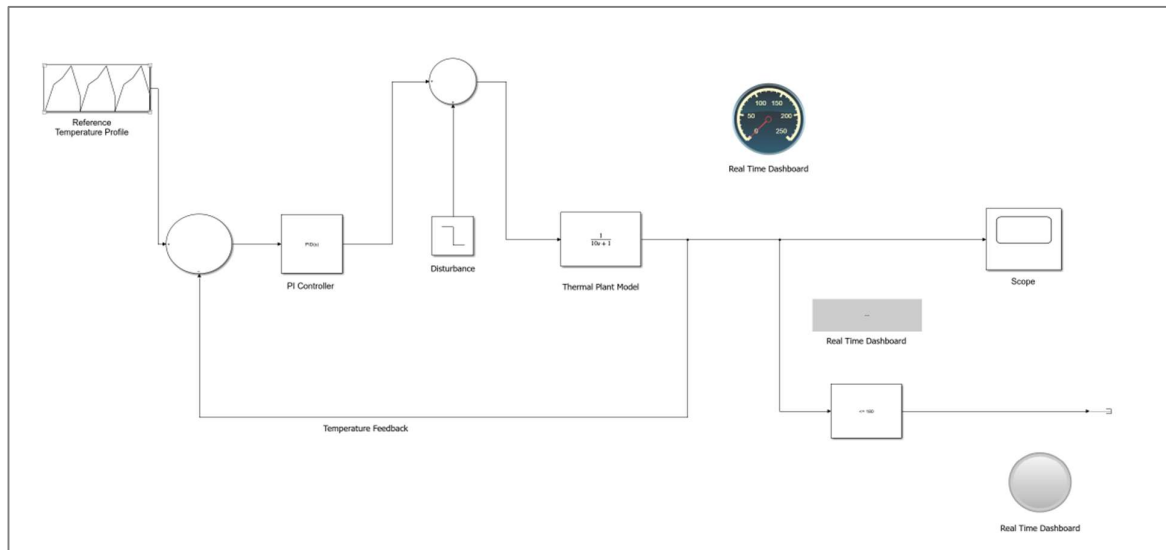


Figure 6.1: Model developed in MATLAB/Simulink.

Table 6.1: Table showing the Simulink Blocks used in model with their corresponding functions.

Simulink Block	Function
Repeating Sequence	Generates the desired reflow temperature profile used as the reference input.
Sum Block	Calculates the error by comparing the reference temperature with the feedback temperature.
PI Controller	Produces the control signal based on proportional and integral actions.
Transfer Function	Represents the thermal dynamics of the reflow oven.

Step Block	Introduces an external disturbance to evaluate controller robustness.
Scope	Displays the system response and temperature tracking graphically.
Circular Gauge	Indicates the current simulated oven temperature.
Numeric Display	Shows the exact numerical value of the temperature.
Indicator Lamp	Indicated when the temperature exceeds the predefined threshold.

6.4 Transfer Function Development

To analyse and control the thermal behaviour of the reflow oven, a mathematical model of the system was required. Since the heating process of a reflow oven exhibits slow and gradual temperature variations, it can be reasonably approximated as a first order dynamic system. This simplified representation captures the essential thermal characteristics of the oven while keeping the controller design and simulation process straightforward.

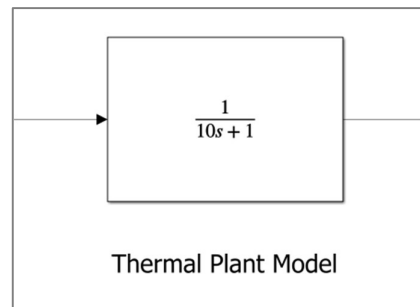


Figure 6.2: Transfer Function Block.

In this project, thermal system was modelled using the standard first order transfer function:

$$G(s) = \frac{K}{\tau s + 1}$$

where,

- K = System gain, representing the relationship between the input heating signal and the output temperature.

- τ = Thermal time constant, representing the speed at which the oven responds to change in the heating input.
- s = Laplace domain variable.

In the model I have used the system gain (K) equals to 1 and Time constant equals to 10 seconds so the transfer function becomes,

$$G(s) = \frac{1}{10s + 1}$$

The first order model assumes that the heating process has a dominant thermal time constant and that higher order thermal effects are negligible for simulation purposes. It provides a good balance between model accuracy and computational simplicity that's why this approximation is commonly adopted in thermal control studies.

Different values of time constants were considered during the simulation for the study of influence of thermal dynamics on controller performance. And the response of PI controller was evaluated for each case, allowing a comparison of system speed, stability and temperature tracking performance.

To evaluate the influence of thermal dynamics on controller performance, simulations were carried out using different values of the thermal time constant. The comparative results are presented in section 7.8.

6.5 PI Controller Design

In this project I have used a Proportional-Integral (PI) controller to regulate the temperature of reflow oven which was implemented in the closed loop control system. The primary objective of the controller is to minimize the error between the desired temperature profile (i.e. reference signal) and the actual oven temperature by continuously adjusting the control input.

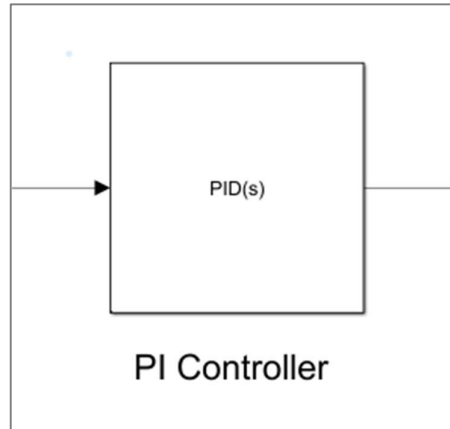


Figure 6.3: PI Controller Block.

The error signal produced at summing point goes through the controller and then controller calculates the appropriate control action using the proportional and integral components. The proportional term provides an immediate response to the current error and improves the speed of system, whereas the integral term accumulates past errors and eliminates steady state error and ensures that the output temperature closely follows the reference profile.

The mathematical expression of the PI controller is given by:

$$C(s) = K_p + \frac{K_i}{s}$$

where,

- K_p = Proportional gain
- K_i = Integral gain
- s = Laplace variable

The PI controller was chosen instead of a full PID controller because the reflow oven behaves as a slow thermal system. In such systems, derivative action generally offers limited improvement and may amplify noise or introduce unnecessary complexity. A PI controller provides a simpler and more robust solution while maintaining satisfactory temperature regulation and eliminating steady state error.

The PI controller was implemented using the PID Controller block in MATLAB/ Simulink which was configured in PI mode, i.e. only the proportional and integral gains were active. The controller output was applied to the thermal model of the reflow oven, forming a closed loop feedback system for continuous temperature regulation.

Parameter	Function
K_p	Increases the speed of response by reacting to the present error.
K_i	Eliminates the steady state error by integrating the accumulated error over time.

After designing the PI controller, the complete control system was implemented in MATLAB/Simulink. The implementation details including reference temperature generation, feedback configuration and monitoring components are presented in next section.

6.6 Temperature Profile Tracking

The reflow soldering process requires the temperature of the Printed Circuit Board (PCB) to follow a carefully controlled thermal profile to ensure proper melting and solidification of the solder paste. Instead of heating the assembly rapidly to the peak temperature, the process is divided into the four distinct stages, each serving a specific purpose.

In the developed MATLAB/Simulink model, the desired temperature profile was generated using the repeating sequence block. This block produces a predefined reference signal representing the complete thermal cycle of the reflow oven. The PI controller continuously compares the actual oven temperature with this reference profile and adjusted the control signal to minimize the tracking error.

Preheating stage

The preheating stage increases the temperature of PCB gradually from the room temperature to approximately 150°C. A controlled heating rate minimizes the thermal shock, activates the solder flux and prepares the assembly for subsequent heating stages.

Soaking Stage

During the soaking stage, the temperature is maintained between 150°C and 180°C for a short duration of time. This allows the entire PCB assembly to reach a uniform temperature and enables the flux to remove surface oxides before solder melting begins.

Reflow Stage

In the reflow stage, the temperature is increased to a peak of approximately 235-250°C, depending the solder alloy used. At this temperature, the paste melts and forms the strong metallurgical bonds between the surface-mounted components and the PCB pads.

Cooling Stage

After soldering, the temperature is allowed to decrease gradually in the cooling stage. Controlled cooling allows the solder joints to solidify properly while minimizing residual thermal stresses and ensuring reliable mechanical and electrical connections.

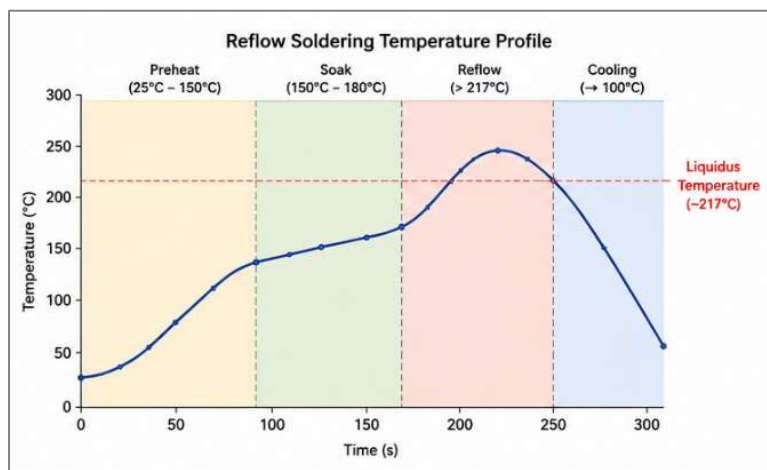


Figure 6.4: Expected Response of MATLAB Simulation.

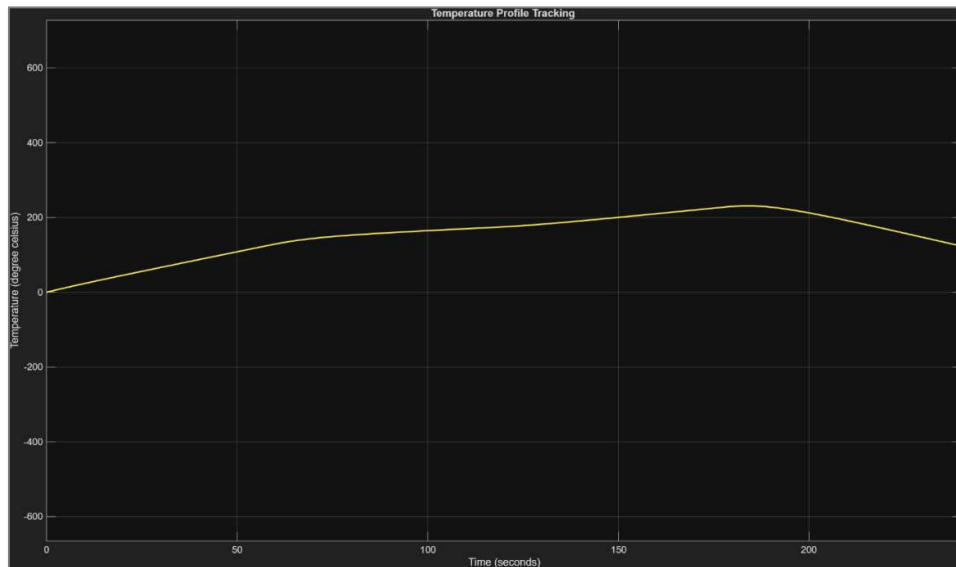


Figure 6.5: Actual Response of MATLAB Simulation.

Following the implementation of the reference temperature profile, the simulated oven temperature was observed to closely follow the desired profile under normal operating conditions. The effectiveness of the tracking process was further evaluated through time constant comparison, disturbance analysis and controller tuning in the subsequent sections.

A summary of the four stages of temperature profile tracking can be observed through the table:

Stage	Temperature Range	Purpose
Preheating	Room temperature – 150°C	Gradual heating and flux activation.
Soaking	150°C – 180°C	Uniform heating and oxide removal.
Reflow	235°C – 250°C	Melting of solder paste and joint formation.
Cooling	Peak temperature to ambient	Controlled solidification of solder joints.

6.7 Time Constant Comparison

To investigate the effects of the thermal dynamics on the performance of the temperature control system, simulation was carried out using three different values of the thermal time constant: $\tau=5s$, $\tau=10s$ and $\tau=20s$. The objective was to understand how changes in time constant affects the thermal response of reflow oven and the how thermal response influences its ability to follow the desired temperature profile.

The time constant determines how quickly the thermal system responds to changes in the control input. A smaller time constant represents a faster response whereas a large time constant represents a slower response. By comparing the responses obtained at different time constants the robustness of the control techniques under varying thermal conditions can be evaluated.

Case 1: Time Constant ($\tau=5s$)

The system was first simulated with a thermal time constant of 5 seconds. Under this condition, the thermal model exhibited the fastest response among all the cases. The output closely

followed the desired temperature profile with a shorter rise time and faster settling thus resulting in excellent tracking performance and minimal delay.

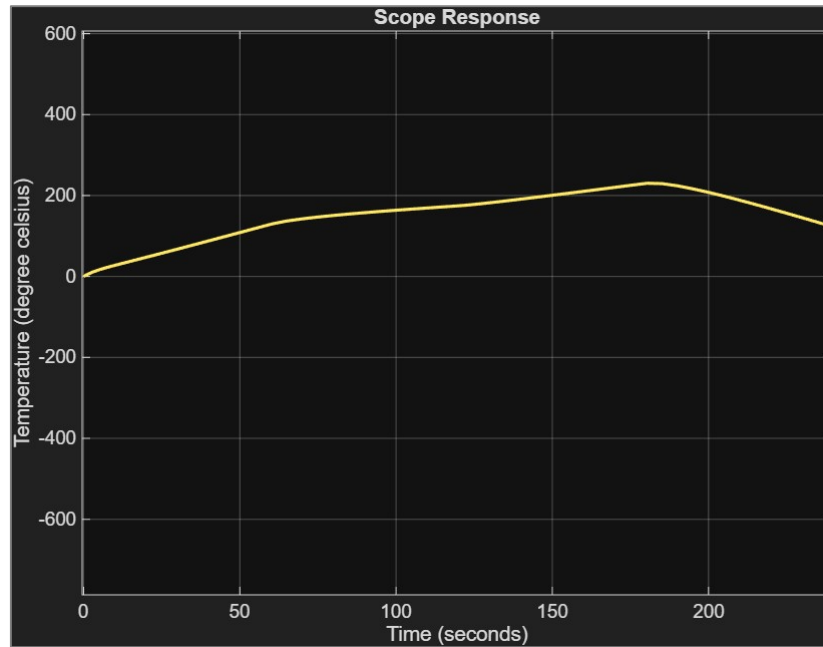


Figure 6.6: Scope Response with time constant 5 seconds.

Case 2: Time Constant ($\tau=10s$)

The model was then simulated with a thermal time constant of 10 seconds. Compared to the previous case, the system responded at moderate speed while maintaining stable operation. The output successfully tracked the reference profile with a slight increase in response time, demonstrating balanced performances between speed and stability.

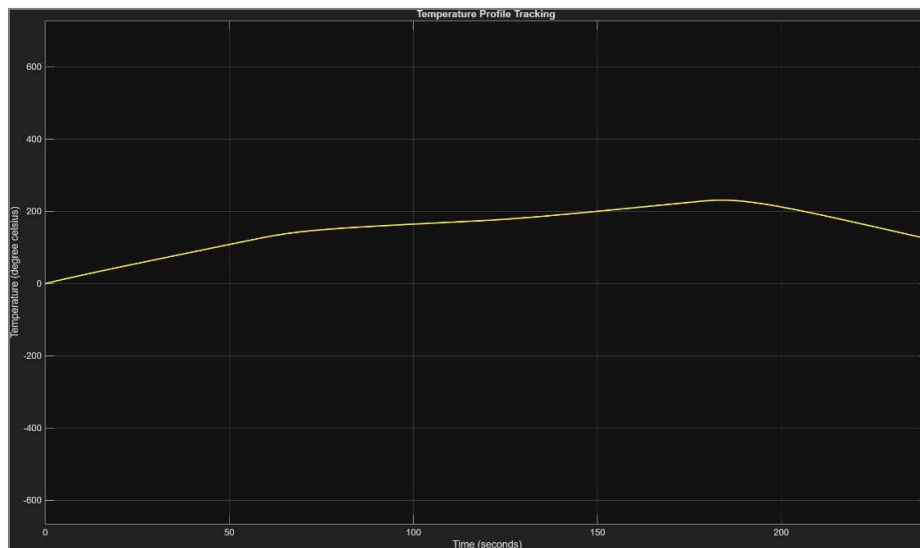


Figure 6.7: Scope Response with time constant 10 seconds.

Case 3: Time Constant ($\tau=20s$)

Finally, the simulation was carried out with a thermal time constant of 20 seconds. Due to a larger thermal inertia, the system exhibited the slowest response. The output required more time to reach the desired temperature, leading to a noticeable delay in tracking the reference profile. However, despite the slower response, the closed loop response remained stable throughout the simulation.

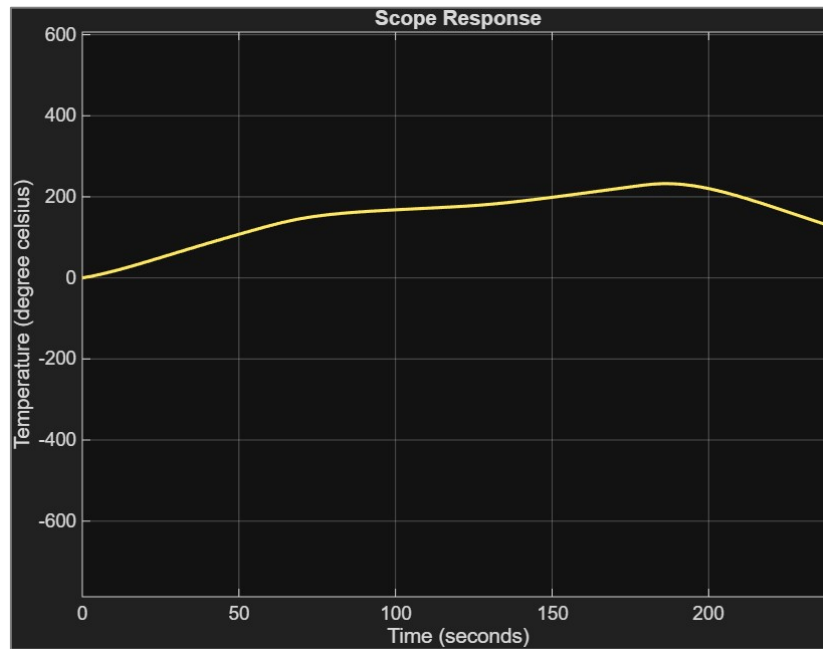


Figure 6.8: Scope Response with time constant 20 seconds.

Comparison and Discussion

The simulation results demonstrate that the thermal time constant has a significant influence on the dynamic response of reflow oven. Although the response speed varied for the varying values of time constant but the PI controlled system remained stable in all the three cases, and this indicates that the proposed control strategy can effectively regulate the oven temperature under different thermal conditions.

Table 6.2: Comparative Study of scope responses at different time constants values.

Parameter	$\tau=5$ sec	$\tau=10$ sec	$\tau=20$ sec
Response Speed	Fastest	Moderate	Slowest

Thermal Realism	Low	High	High
Tracking Performance	Excellent	Good	Acceptable
Suitability for Further Analysis	Low	Selected	Low
System Stability	Stable	Stable	Stable

On the basis of comparative analysis, thermal constant of 10 seconds was selected for the subsequent stages of the project. Although the system with time constant 5 seconds has the fastest response that means a system with very low thermal inertia, which is generally less representative of the heating characteristics of practical industrial reflow ovens. On the other hand, time constant 20 seconds generating a considerably slower response.

The time constant 10 seconds provided a balanced response by offering satisfactory tracking performance while maintaining realistic thermal behaviour. It closely represents the gradual heating characteristics expected in industrial thermal process and therefore served as suitable operating condition for the remaining analyses, including disturbance testing, Pi controller tuning and dashboard implementation.

6.8 Disturbance Analysis

If we consider a practical industrial application, a reflow oven is subjected to various disturbances that can affect its thermal performance. Those disturbances can be arisen due to variations in ambient temperature, fluctuations in heater output, changes in PCB loading or frequent opening of the oven chamber. That's why it becomes necessary to evaluate the model's ability to maintain a stable temperature under the disturbed operating conditions.

To make these conditions into considerations, a step block was introduced into the Simulink model. The disturbance was applied during the simulation to represent sudden change in the thermal system. The main objective was to analyse the response of PI controlled system and evaluate its capability to reject the disturbance to maintain the desired temperature profile.

Case 1: Positive Disturbance

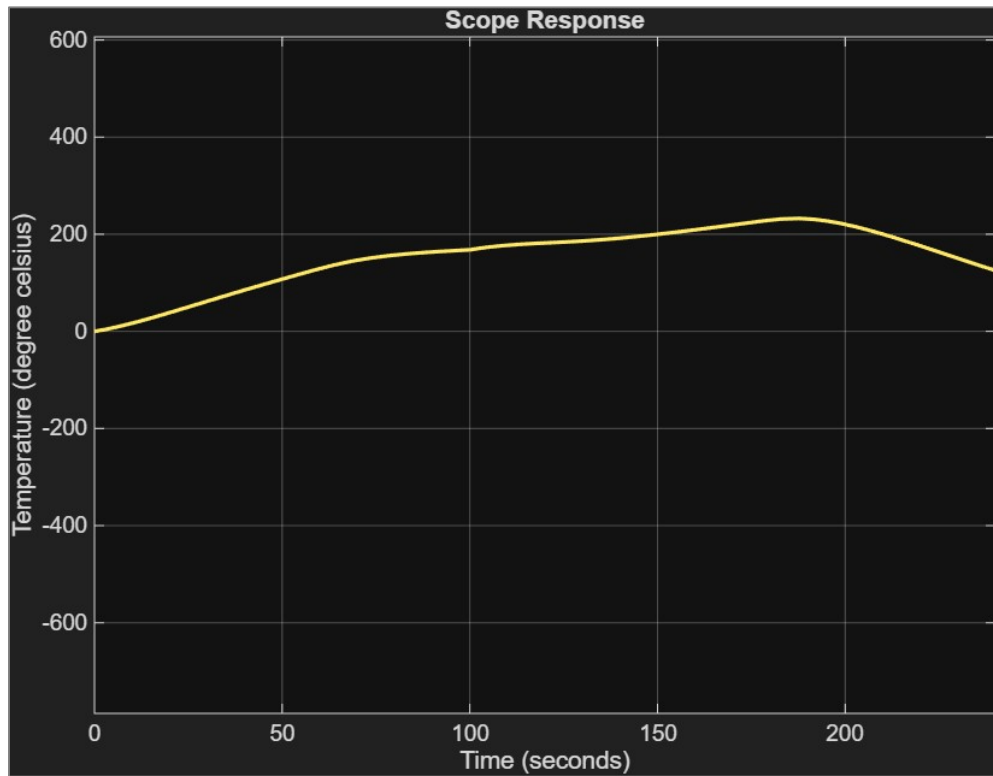


Figure 6.9: Scope Response with Positive Disturbance.

A positive disturbance of maxima 20 was applied at the 100th second of simulation. An unexpected increase in the system output was observed at the 100th second. However, the PI controller responded by adjusting the control signal and gradually restored the temperature to its desired value. The system remained stable throughout the simulation and recovered without sustained oscillations.

Case 2: Negative Disturbance

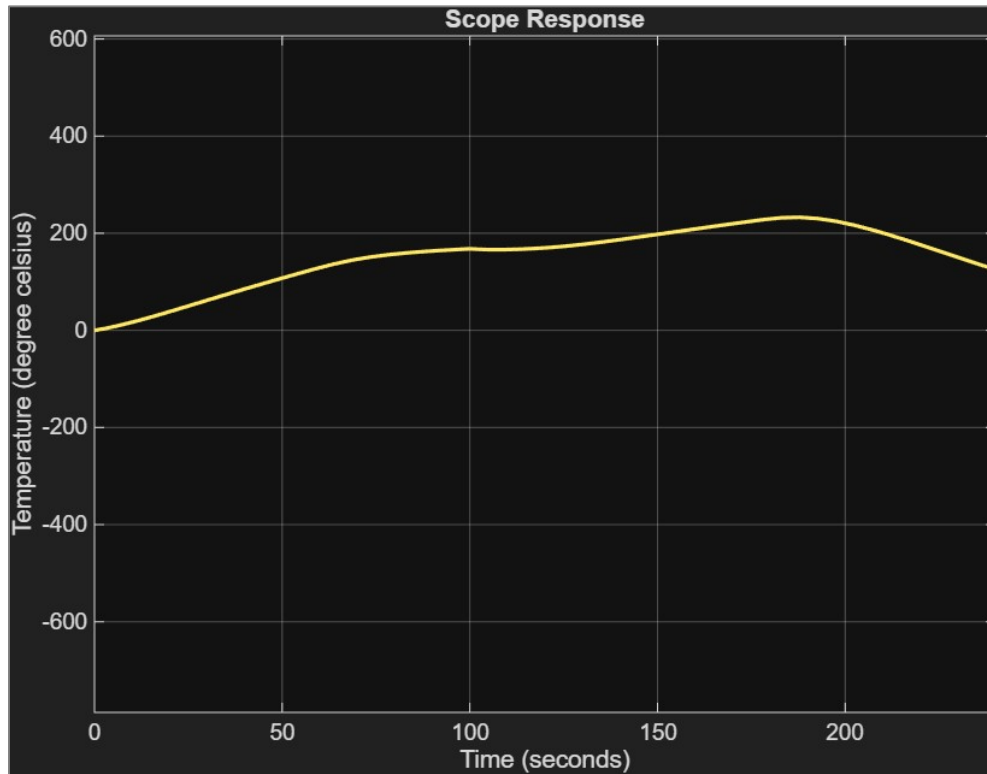


Figure 6.9: Scope Response with Negative Disturbance.

A negative step disturbance of step size -20 was applied to represent a sudden reduction in the system input. As expected, the temperature output decreased below the reference value. The PI controller again compensated this deviation by adjusting the control signal. The system remained stable, demonstrating the controller's ability to handle both positive and negative disturbances.

Although both positive and negative disturbances can be used to evaluate the robustness of the temperature control system. A positive disturbance represents an unexpected rise in heat generation whereas negative disturbance represents the sudden reduction in heat input such as heat loss due to the opening of oven door or introduction of a colder PCB into the reflow oven.

In this project, the negative disturbance was applied because it is the one of the most common and practically significant disturbances happens in the industrial systems. Moreover, recovering from heat loss is more challenging for a controller rather than responding to a temporary increase in the temperature as the controller have to supply additional heat energy to maintain the desired temperature.

Also, a positive disturbance could have been used but, since the negative disturbance provides a more realistic assessment to controller's rejection capability under typical operating conditions. And the successful recovery of the system following the applied -20 disturbance demonstrates the effectiveness and robustness of the proposed PI controller in maintaining the desired temperature profile.

6.9 Manual PI Tuning

Before applying MATLAB's automatic tuning technique, the PI controller was tuned manually to understand the influence of the proportional and integral gains on the temperature response of reflow oven. Manual tuning was carried out by trial-and-error method in which controller gains were adjusted iteratively while observing the system response.

Initially the controller gains were set to the values such as, $K_p = 1$ and $K_i = 0.1$. And the corresponding result was stable but relatively slow, with noticeable tracking delay when the reference temperature profile was changed. To improve system performance, the controller parameters were adjusted while observing the results after each simulation.

After several iterations, the controller gains were manually tuned to , $K_p = 2$ and $K_i = 0.2$. These values produce a faster response comparatively and improved tracking performance while maintaining the stable operation. The response time was reduced and there was a satisfactory temperature regulation compared to the initial parameters.

Although the manual tuning process resulted in improved controller performance but it required repeated simulations and depended on engineering judgment. To obtain a more systematic and optimized set of controller parameters, MATLAB's `pdtune()` function was used. The automatically tuned controller is discussed in the following section.

The response of manual tuning in both the cases can observed and compared from the given responses below:

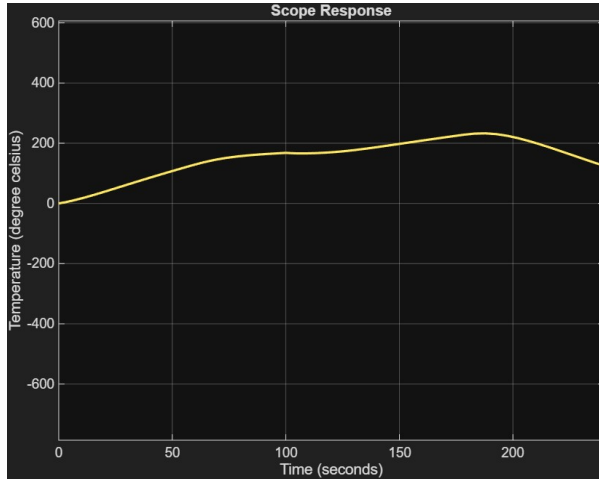


Figure 6.10: Scope Response with $K_p=1$ and $K_i=0.1$.

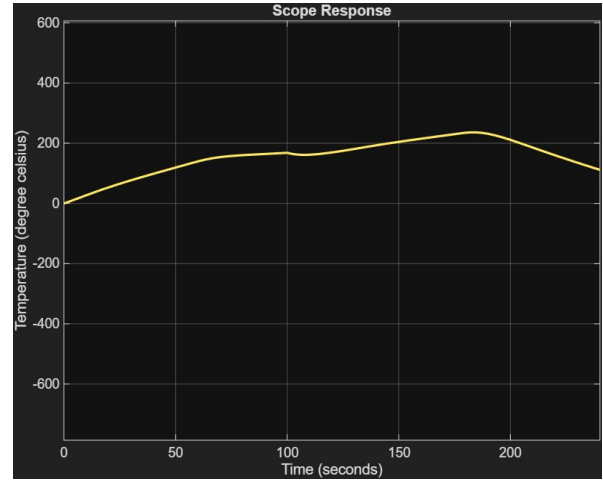


Figure 6.11: Scope Response with $K_p= 2$ and $K_i=0.2$.

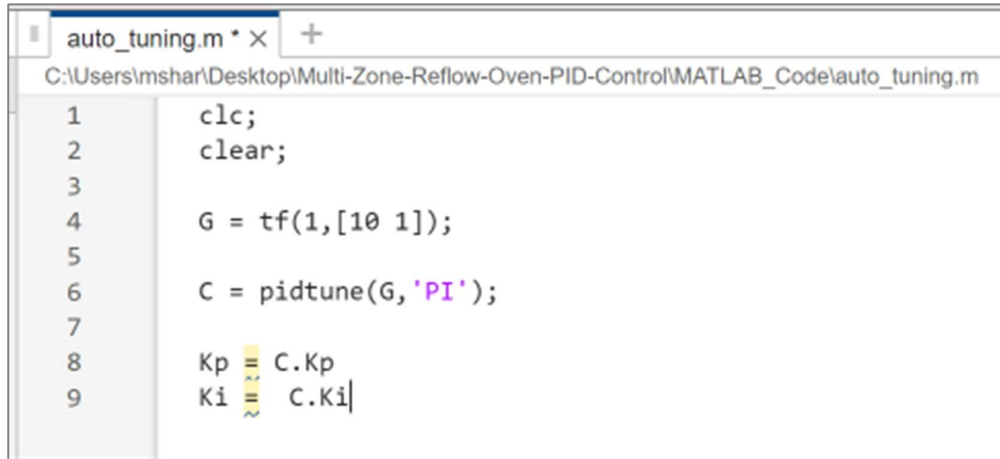
The comparison between the initial and manually tuned controller demonstrates that increasing both the proportional and integral gains improved the responsiveness of the system without compromising stability. The manual tuning process provided valuable insight into the influence of controller parameters on the thermal response. However, because trial-and-error tuning can be time-consuming and may not always yield optimal results, automatic tuning using MATLAB's `pidentune()` function was performed to obtain improved controller parameters.

6.10 Automatic PI Tuning using `pidentune()`

Manual tuning produced a satisfactory result but the tuning process required a number of iterations and was relied on trial-and-error method. Thus, to obtain a more systematic and optimized set of controller parameters, MATLAB's `pidentune()` function was used.

The `pidentune()` function automatically determines the suitable controller gains based on the mathematical model of the plant. It analyses the transfer function of the thermal system and computes the controller parameters that provide a good balance between response speed, stability and robustness. This approach reduces the time required for controller design while improving tuning accuracy.

In this project, the first order transfer function was supplied to `pidentune()` function to design a PI controller. The MATLAB code used for tuning is shown below:



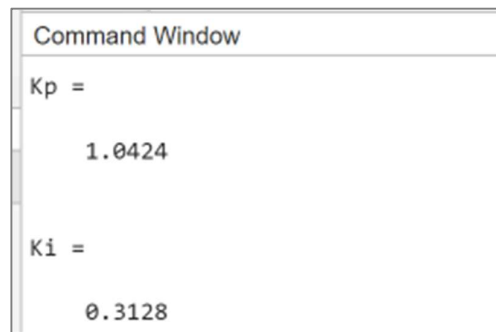
```
1  clc;
2  clear;
3
4  G = tf(1,[10 1]);
5
6  C = pidtune(G,'PI');
7
8  Kp = C.Kp
9  Ki = C.Ki
```

Figure 6.12: MATLAB Code for Automatic Tuning.

The automatically tuned controller produced the following results:

- Proportional Gain (K_p) = 1.0424
- Integral Gain (K_i) = 0.3128

These gains were then implemented in the Simulink model, and the system response was evaluated. The tuned PI controller produced the improved temperature tracking with response and reduced the effort associated with manual parameter adjustment.



```
Command Window

Kp =

    1.0424

Ki =

    0.3128
```

Figure 6.13: Controller Parameters determined by Automatic Tuning.

Both manual and automatic tuning produced stable controller performance. Manual tuning provided valuable insight into the influence of controller gains on the system response, whereas the `pidtune()` function offered a faster and more systematic approach to controller design. The automatically tuned controller achieved improved tracking performance while reducing the time and effort required for parameter selection. Therefore, the gains obtained using `pidtune()` were adopted for the final implementation of the temperature control system.

6.11 Dashboard Implementation

For improving the visualization and monitoring of the simulated temperature control system, an interactive dashboard was built using the dashboard blocks available in MATLAB/Simulink. Dashboard provides real time visualizations of the system response thus allowing the user to monitor the simulated oven temperature and operating status during the simulation.

The dashboard consists of following components:

Circular Gauge

The circular gauge represents the simulated instantaneous temperature of reflow oven graphically, and enables the quick observation of temperature variations throughout the heating cycle.

Numeric Display

The numeric Display displays the exact numerical value of oven temperature during the simulation thus allowing precise monitoring of the system response.

Constant Comparator

The Constant Comparator block compares the simulated oven temperature with a predefined threshold value. When temperature exceeds the threshold value, the comparators produce logical output (by default TRUE) that is used to indicate operating status of system. In this project threshold value is defined as 180.

Indicator Lamp

The indicator lamp is connected to the output of constant comparator. The lamp glows green when the temperature exceeds the threshold value 180 and remains grey coloured when temperature is below the threshold value.

A threshold value of 180°C was selected for the constant comparator because it represents the transition from the soaking stage to the reflow stage of the temperature profile. This is the stage where PCB assembly has completed its uniform heating and is approaching the temperature at which solder paste begins to melt. 180°C is used as the threshold because it is

the meaningful indication that the oven has entered the critical heating phase of the reflow process. And thus, indicator lamp changes its state to notify the user that the desired operating region has been reached.

6.12 Simulation Results and Discussion

The proposed PI controlled reflow oven temperature control system was successfully modelled and simulated in MATLAB/Simulink. Simulation results demonstrates that the developed controller effectively regulated the oven temperature and closely tracked the desired reflow temperature profile under different applied conditions.

Initially the thermal characteristics were analysed by comparing the model at different time constants that is 5s, 10s and 20s. The study suggested that a smaller time constant results in a faster response whereas a large time constant has a slow response due its increased thermal inertia. Thus, based on the overall performance and practical thermal behaviour time constant 10s was selected for the further analyses.

To evaluate the robustness of the PI Controller a disturbance of -20 was introduced in to the model. The disturbance caused the deviation in the output temperature but the PI controller successfully compensated the disturbance and restored the system to the desired operating condition without instability.

Performance of controller was further investigated by tuning the PI controller through both manual and automatic tuning methods. Manual tuning improved the response by adjusting the controller parameters from $K_p = 1$, $K_i = 0.1$ to $K_p = 2$, $K_i = 0.2$. After that controller parameters were obtained through the MATLAB's automatic tuning function `pdtune()` and the parameters were found to be $K_p = 1.0424$ and $K_i = 0.3128$. And these parameters were employed to the system for further investigation.

An interactive dashboard was also developed using the Simulink dashboard blocks which includes the circular gauge, numeric display, constant comparator and the indicator lamp. The dashboard enabled the user with an improved visualizations and interpretations of simulation results.

Fig.6.14 presents the final simulation response of the proposed PI controlled reflow oven using the selected parameters. The output temperature closely follows the desired reflow temperature profile throughout the simulation cycle. The response is stable with satisfactory

tracking performance and maintains desired temperature trajectory. The graph also shows that after the introduction of the negative disturbance; the controller effectively compensates for the deviation and restores the system to the desired operating condition without instability. The overall simulation result demonstrates the effectiveness of the proposed control strategy for regulating the thermal process of the reflow oven.

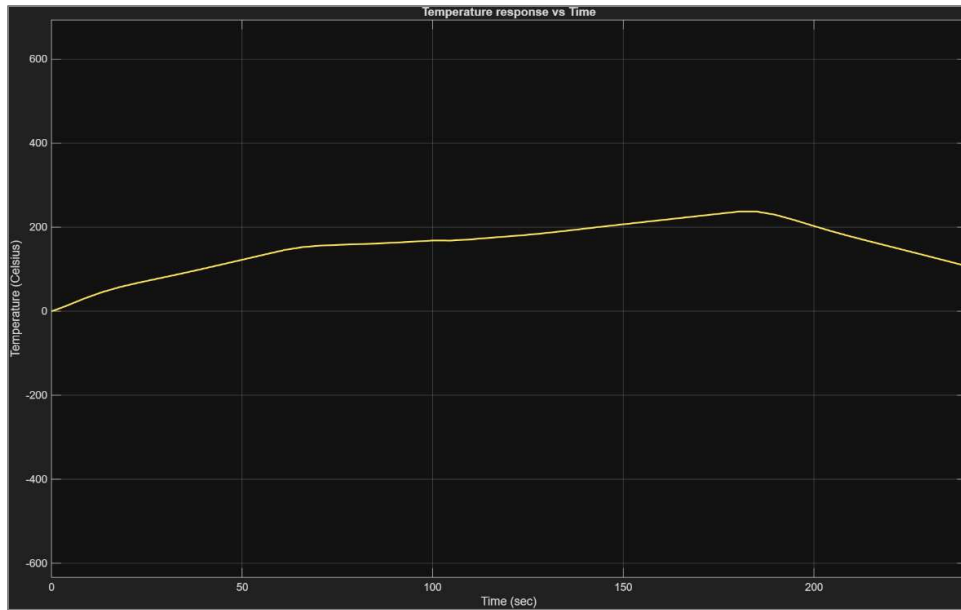


Figure 6.14: Final Scope Response of Simulation.

Overall, the simulation demonstrates that the PI control strategy provides a stable temperature regulation, satisfactory tracking performance and effective disturbance rejection making it suitable for temperature control applications in reflow soldering systems.

7. Challenges Faced

Several technical challenges were faced during the project development and simulating the PI controlled reflow oven temperature control system. And these challenges provide a valuable learning opportunity and a practical understanding of MATLAB/Simulink and control system design.

- One of the challenges was to develop the mathematical model of the reflow oven as the no real industrial data was available, thus a simplified first order transfer function was chosen based on thermal characteristics of the reflow oven.
- To determine a suitable time constant multiple simulations had to be carried out. Different values of time constants were applied and response was analysed then a final value for further investigation was chosen.
- For a good performance of PI controller, the tuning was done manually first by iterating the controller parameters with different values. Understanding the influence of changing the gain values was essential to understand for the controller design.
- Integrating the dashboard blocks with Simulink model required proper signal connections. Ensuring that each component displayed correct simulation data was an important part of implementation.
- Preparing a well-structured technical report required organizing simulation results, figures, tables, and explanations in a logical sequence while avoiding unnecessary repetition across different sections.

Outcome

The challenges encountered during the internship strengthened practical skills in thermal system modelling, controller design, MATLAB/Simulink simulation, troubleshooting, and technical documentation. Successfully overcoming these challenges contributed significantly to the completion of the project.

8. Conclusion

The objective of this internship project was to develop and simulate a PI controlled temperature control system for a multizone reflow oven using MATLAB/Simulink. Thus, a mathematical model of the thermal system was developed and a PI controller was designed to regulate the oven temperature according to the required reflow temperature profile.

The completed simulation demonstrated that the proposed control strategy is capable of achieving stable temperature regulation and satisfactory tracking of the desired thermal profile. The project also highlighted the importance of proper controller tuning and process modelling in achieving reliable thermal control. An interactive dashboard was developed to provide real-time monitoring of the simulated process, making the system easier to observe and analyse.

Through this project, practical knowledge of thermal process modelling, transfer function development, PI controller design, MATLAB programming, Simulink simulation, controller tuning, and industrial process visualization was gained. The internship provided valuable exposure to control engineering concepts and strengthened the ability to apply theoretical knowledge to an industrially relevant application.

Overall, the project successfully met its objectives and demonstrated the effectiveness of PI control for temperature regulation in a simulated reflow oven environment.

9.Future Scope

Although the proposed PI controlled reflow oven temperature control system showed satisfactory results in simulation but several enhancements can be made in future to improve its functionality and practical applicability.

Some possible future developments are as follows:

1. Real Time Hardware Implementation

This proposed controller can be implemented on an actual hardware using microcontrollers, PLCs or embedded control platforms to analyse the simulation results under real operating conditions.

2. Multizone Temperature Control

Present work includes a simplified thermal model but in future separate controllers can be designed for the different heating zones. It will help to achieve more accurate temperature control throughout the reflow oven.

3. Advanced Control Techniques

More advanced techniques such as Model Predictive Control (MPC), Fuzzy Logic Control or adaptive Control can be investigated and compared with the PI controller to have an improved performance control system.

4. Integration with IOT

The temperature monitoring system can be integrated with IOT technology for remote supervision, real time data logging and predictive maintenance of industrial reflow ovens.

5. AI based Optimization

Machine learning and Artificial Intelligence Techniques can be used to automatically optimize controller parameters, to predict temperature variations and to improve energy efficiency.

6. Industrial Validation

This proposed technique can also be tested on industrial reflow oven to compare the simulated performance with a real manufacturing condition and evaluate its practical effectiveness.

References

1. K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton, NJ, USA: Princeton University Press, 2008.
2. K. Ogata, *Modern Control Engineering*, 5th ed. Upper Saddle River, NJ, USA: Pearson Education, 2010.
3. N. S. Nise, *Control Systems Engineering*, 8th ed. Hoboken, NJ, USA: John Wiley & Sons, 2019.
4. MathWorks, **Control System Toolbox Documentation**. Available: <https://www.mathworks.com/help/control/>
5. MathWorks, **Simulink Documentation**. Available: <https://www.mathworks.com/help/simulink/>
6. MathWorks, **pidtune Documentation**. Available: <https://www.mathworks.com/help/control/ref/dynamicsystem.pidtune.html>
7. MathWorks, **Transfer Function (tf) Documentation**. Available: <https://www.mathworks.com/help/control/ref/tf.html>
8. J. H. Lau, *Reflow Soldering Processes and Troubleshooting: SMT, BGA, CSP and Flip Chip Technologies*. New York, NY, USA: McGraw-Hill, 2003.
9. S. Kalpakjian and S. R. Schmid, *Manufacturing Engineering and Technology*, 7th ed. Pearson Education, 2014.
10. IPC Association Connecting Electronics Industries, *IPC-7530: Guidelines for Temperature Profiling for Mass Soldering Processes (Reflow and Wave)*, Bannockburn, IL, USA.

Internship Certificate



YOUR PARTNER
BEYOND PRODUCTS

ELIN/GZB/ 393 /26-27

Date- 22.07.2026

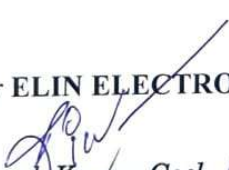
TO WHOM SO EVER IT MAY CONCERN

This is to certify that **Ms. Preeti Sharma D/o Shri Mahesh Sharma (Roll no 24106069)**, a Second-year student of **Instrumentation And Control Engineering at Dr. B.R. Ambedkar National Institute of Technology, Jalandhar** has successfully completed her industrial training in the Engineering Department of our organization from **June 1, 2026 to July 22, 2026**.

During her internship tenure, we found her to be **sincere, dedicated, and enthusiastic**. Her performance throughout the training period was good.

We wish her all the very best and every success in her future career endeavors.

For **ELIN ELECTRONICS LTD.,**


Dinesh Kumar Goel
HR Head



ELIN ELECTRONICS LIMITED

CIN: L29304DL1982PLC428372

New Delhi

Registered & Corporate Office
4771, Bharat Ram Road,
23, Daryaganj,
New Delhi-110002

☎ 011-43000400

Ghaziabad

C-142,143,144,144/1,144/2
Bulandshahr Road
Industrial Area, Site No. 1,
Ghaziabad (U.P.)-201009

☎ 0120-2701519

Goa

L-84, Verna Industrial
Estate Verna, Salcete
Goa-403722

☎ 08326690939

Baddi

Village-Belikhoh,
Tehsil-Nalagarh,
District: Solan,
Himachal Pradesh-174101

☎ 9816036987

Bhiwadi

C2-C3 & C5-C6
Elcina Manufacturing
Cluster SPL-1, Salarpur
Industrial Area Bhiwadi,
District: Alwar
Rajasthan-301019